TEXORA · LECTURE SHEET

Physical Chemistry of Dyeing

Dyeing is a partition equilibrium approached through diffusion. Affinity says where the dye wants to be; isotherms describe how it distributes; kinetics says how fast it gets there. This sheet derives all three, with the numbers worked.

WP-4101 · TEXTILE CHEMISTRY & COLORATION-III · TOPIC 1 · SPECIALIZATION — WET PROCESSING

1. The dyeing system as a physico-chemical system

A dyeing consists of dye dissolved (or dispersed) in the bath, the fibre as a second phase, and the auxiliaries that mediate between them. Dye transfers from bath to fibre until the chemical potentials equalise — **equilibrium** — and the speed of transfer is governed by diffusion through the bath boundary layer and inside the fibre. Everything practical in dyeing — salt, temperature, time, levelling agents — is an intervention in this equilibrium or this diffusion. (Reactive dyes add a chemical reaction on top of the physical picture; the physical stage still delivers the dye to the fibre first, WP-4101 Topic 2.)

2. Adsorption isotherms — the shape of the equilibrium

An **isotherm** plots dye on fibre $[D]_f$ against dye remaining in solution $[D]_s$ at equilibrium, at constant temperature. Three idealised forms cover the dye classes:

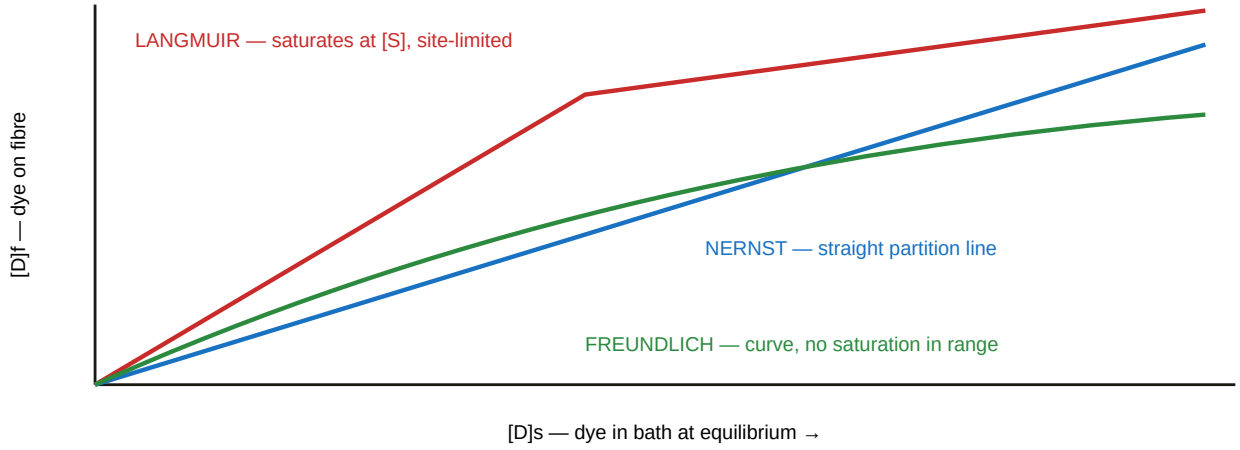


Fig. 1 — The three adsorption isotherm types. The shape is diagnostic: a plateau means finite dye sites; a straight line means simple partition between two phases; a continually rising curve means multilayer-type uptake without a site limit in the practical range.

- **Nernst (partition):** $[D]_f = K\,[D]_s$ — the dye distributes between fibre and bath in a constant ratio, like a solute between two solvents. Disperse dyes in polyester behave this way: the fibre is effectively a solid solution.
- **Freundlich:** $[D]_f = K\,[D]_s^{x}$ $(x < 1)$ — uptake rises steeply at first, then ever more slowly, without a clear plateau; typical of dyes on cellulose (direct, reactive before fixation, vat leuco).
- **Langmuir:** $[D]_f = K\,[D]_s\,[S] / (1 + K\,[D]_s)$ — adsorption onto a fixed number of sites [S]; at high $[D]_s$ the fibre saturates. Acid dyes on the protonated amino sites of wool and nylon are the classical case — and the reason two acid dyes compete for nylon and can block each other.

3. Affinity and the standard free energy

The **affinity** of a dye is its standard chemical potential difference between bath and fibre, $-\Delta\mu^\circ$: the driving force of dyeing. From the equilibrium distribution:

$$-\Delta\mu^\circ = RT \ln\left([D]_f / [D]_s \right)$$

with concentrations in consistent units and $R = 8.314$ J/mol·K.

> **WORKED EXAMPLE 1 — AFFINITY FROM AN EQUILIBRIUM DISTRIBUTION**
>
> At 80 °C (353 K) a dye reaches equilibrium with $[D]_f = 25$ g per kg of fibre and $[D]_s = 0.25$ g per kg of bath — a distribution ratio of 100.
>
> $-\Delta\mu^\circ = 8.314 \times 353 \times \ln(100) = 8.314 \times 353 \times 4.605 =$ **13.5 kJ/mol**.
>
> A ratio of 100 at 353 K is a moderate affinity; direct dyes on cellulose show values of this order, while acid dyes on wool run higher. Affinity is a property of dye + fibre + temperature — quote all three.

4. Heat of dyeing and entropy — the van't Hoff analysis

Measuring the distribution at two temperatures separates affinity into its enthalpy and entropy parts ($\Delta G = \Delta H - T\Delta S$):

> **WORKED EXAMPLE 2 — ΔH AND ΔS FROM TWO TEMPERATURES**
>
> The dye of Worked Example 1 gives distribution ratio $K = 100$ at 353 K but only $K = 60$ at 373 K (100 °C) — exhaustion falls as temperature rises, the usual behaviour.
>
> van't Hoff: $\ln(K_2/K_1) = -(\Delta H/R)(1/T_2 - 1/T_1)$. $\ln(60/100) = -0.511$; $(1/373 - 1/353) = -1.519 \times 10^{-4}$ K^{-1}.
>
> $\Delta H = -R \times (-0.511)/(-1.519 \times 10^{-4}) =$ **−28.0 kJ/mol** — dyeing is exothermic: heat is released when dye adsorbs, so raising temperature drives dye back into the bath (Le Chatelier, in numbers).

$\Delta S = (\Delta H - \Delta G)/T = (-27{,}960 + 13{,}515)/353 =$ **−40.9 J/mol·K** — negative: the dye is more ordered adsorbed on sites in the fibre than free in solution. Dyeing proceeds because the enthalpy gain outweighs the entropy loss — and stops (equilibrium) exactly where the two balance.

Practical consequences: (i) final exhaustion of most dye–fibre systems is higher at lower temperature — but (ii) diffusion is faster hot, so processes dye hot for rate and levelness and accept the equilibrium penalty, or dye at the boil and add salt to recover exhaustion. Temperature programming in machines (WP-3207 Topic 7) is this thermodynamics made operational.

5. Exhaustion — the production measure of equilibrium

$$\% \text{ Exhaustion} = (\text{initial bath dye} - \text{equilibrium bath dye}) / \text{initial bath dye} \times 100$$

WORKED EXAMPLE 3 — EXHAUSTION AND ON-FIBRE DEPTH

1 kg of cotton at LR 1:20 (20 L bath). Initial dye concentration 0.60 g/L; at equilibrium 0.15 g/L.
Exhaustion = (0.60 − 0.15)/0.60 × 100 = **75%**.
Dye on fibre = (0.60 − 0.15) g/L × 20 L = **9.0 g** → 0.90% owf on the fibre (from a nominal 1.20% owf in the bath).
The 0.30% owf left in the bath is not a rounding error: it is effluent colour and lost money, and it is why exhaustion appears in every serious recipe review.

6. Kinetics — diffusion, rate and the half-dyeing time

Equilibrium says where dyeing ends; **kinetics** says whether the lot is level when it gets there. Dye must cross the stagnant boundary layer at the fibre surface (reduced by circulation) and diffuse within the fibre, following Fick's laws with a diffusion coefficient that rises steeply with temperature and with fibre swelling. Early in dyeing, uptake grows roughly with √t (the signature of diffusion control). The **half-dyeing time $t_{½}$** — time to reach half the equilibrium uptake — is the practical rate constant: dyes with similar $t_{½}$ are compatible in mixture dyeing; a fast "striking" dye levels badly because it adsorbs before it can distribute. Levelling agents work kinetically: they complex the dye in the bath or block fibre sites temporarily, slowing adsorption so diffusion can keep the distribution uniform.

7. Ionic fibres and the Donnan equilibrium — why salt works

Cellulose in water carries a negative surface charge; anionic dyes (direct, reactive) are repelled from it. The Donnan membrane equilibrium describes the resulting unequal distribution of mobile ions between fibre phase and bath. Adding electrolyte (NaCl, Na_2SO_4) floods the bath with Na^+, which accumulates

near the fibre and screens the repulsion — dye anions can approach and adsorb, and exhaustion rises. The same theory, reversed, explains why acid dyes strike fast onto positively charged wool at low pH and why Glauber's salt *retards* acid dyeing of wool (there the salt anion competes)... precisely: electrolyte effects are charge effects, and the sign depends on the dye–fibre charge pair. This is the physical chemistry behind every salt figure in Topics 1–5 of WP-3207.

8. Glossary — key terms

- **Affinity (−Δμ°):** standard free-energy driving force for dye transfer to the fibre.
- **Isotherm:** the equilibrium relation between dye on fibre and dye in bath at fixed temperature.
- **Exhaustion:** percentage of bath dye transferred to the fibre.
- **Half-dyeing time:** time to reach half of equilibrium uptake; the practical rate measure.
- **Strike:** the initial rapid adsorption of dye, dangerous to levelness when uncontrolled.
- **Donnan equilibrium:** unequal ion distribution between a charged fibre phase and the bath.

9. Key points

- Isotherm shape identifies the mechanism: partition (Nernst), site saturation (Langmuir), or Freundlich behaviour.
- Affinity from the distribution ratio: 100 at 353 K ⇔ 13.5 kJ/mol; dyeing is typically exothermic (ΔH ≈ −28 kJ/mol) with negative entropy of adsorption.
- Processes are compromises between equilibrium (favours cool) and kinetics (favours hot) — temperature and salt are the levers, and both are now explicable, not traditional.

10. Summary

Physical chemistry converts dyeing from recipe-following into engineering: the isotherm predicts exhaustion, affinity quantifies the driving force, van't Hoff splits it into heat and order, and diffusion sets the timetable on which levelness is won or lost. Every later topic in this course — dye chemistry, processes, faults — is an application of this sheet.

11. Practice

1. A dye shows distribution ratio 250 at 60 °C (333 K). Calculate its affinity −Δμ°.
2. A bath at LR 1:15 starts at 0.80 g/L and equilibrates at 0.28 g/L for a 2 kg lot. Calculate % exhaustion and the dye on fibre in % owf.

3. Which isotherm type would you expect for (a) a disperse dye on polyester, (b) an acid dye on nylon approaching saturation, (c) a direct dye on cotton? Justify each in one line.
4. A dye's exhaustion falls when the bath is heated, yet the process is run at the boil. Resolve the apparent contradiction in two sentences.

Answer key

1. $-\Delta\mu^\circ = 8.314 \times 333 \times \ln(250) = 8.314 \times 333 \times 5.521 = 15{,}287$ J/mol ≈ 15.3 kJ/mol.
2. Exhaustion = (0.80 − 0.28)/0.80 = 65%. Bath volume = 2 × 15 = 30 L; dye on fibre = 0.52 × 30 = 15.6 g on 2,000 g = 0.78% owf.
3. (a) Nernst — the fibre acts as a solvent phase, partition without sites. (b) Langmuir — adsorption onto a fixed number of protonated amino sites, saturating. (c) Freundlich — uptake rising without a practical plateau.
4. Equilibrium exhaustion is indeed lower hot (dyeing is exothermic), but diffusion and levelling are far faster at the boil, and production values level, rapid dyeing over the last points of exhaustion — which salt addition or a cooling hold can partly recover. The boil is a kinetic choice paid for in equilibrium currency.

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